

Learning Outcomes

1. Appraise the development of the human resource function.
2. Critically examine the traditional and contemporary approaches of recruitment and selection.
3. Analyse the procedures and practices of performance management and remedial activities.
4. Investigate and plan the resources required to achieve an organisation's aims and objectives.
- 5. Evaluate and appraise the use of resources against organisational plans.**
- 6. Formulate plans to ensure that resources are maximised and remedial actions taken where appropriate**

Discussions & Methodologies

- Lean Production
- TQM
- Kanban system
- Kaizen – Continuous Improvement

Planning

Organising

Supervising

Efficiency

Resource
Planning

Transformation
Process

Supply chain
management

Operations Management

- Production
- Service delivery

Logistics

Quality

Lead times

Layout

Absenteeism

Staff
turnover

Capacity
Utilisation

Main activities of operations

- “Providing good and services depends on 5 key operations activities and these provide a useful way of describing and analysing an organisations operations systems” (Sprage 2007)
- Capacity
- Standards
- Materials - JIT
- Scheduling – matching supply with demand
- Control – are plans working? (objectives, measuring, comparing & acting)

Operations processes: Production systems

- Job shop systems
- Batch operations
- Assembly Line
- Mass production and continuous flow manufacturing

Product Process Matrix

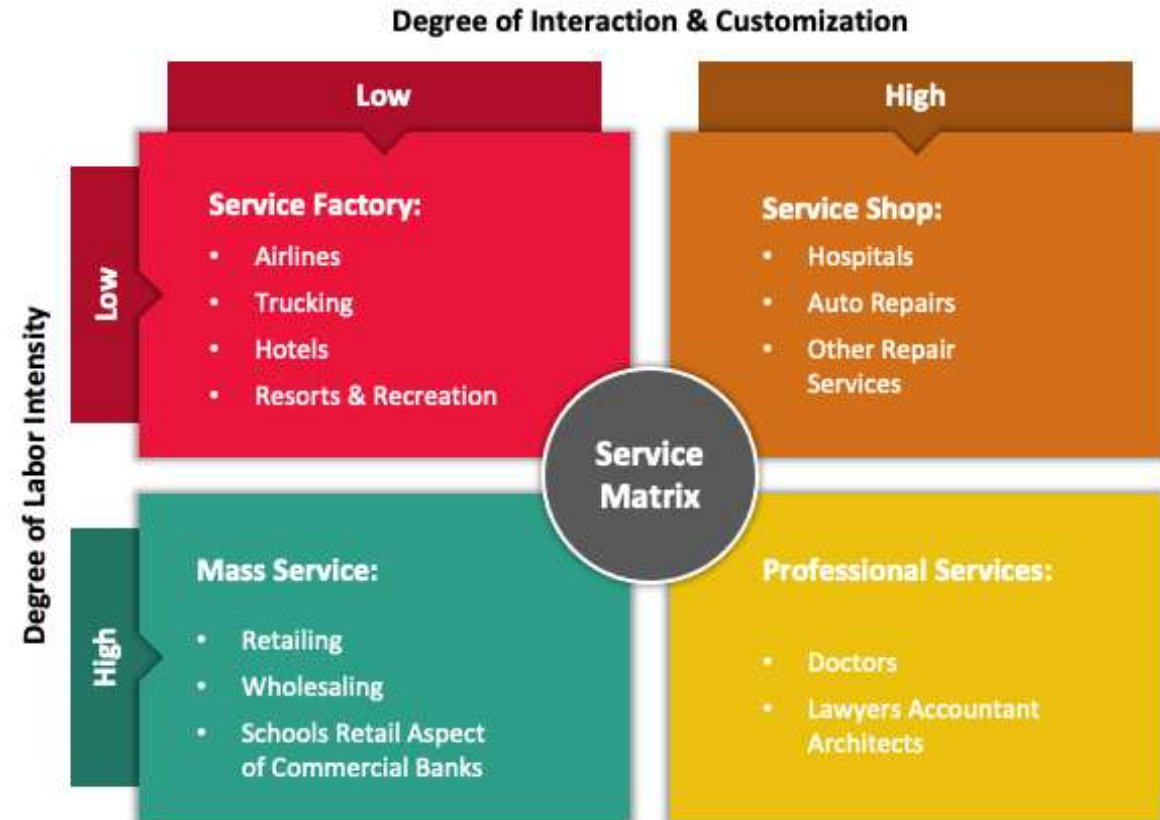
		PRODUCT STRUCTURE			
		Low Volume Unique	Low Volume Multiple Products	Higher Volume Standardized Product	Very High Volume Commodity Product
PROCESS STRUCTURE	Jumbled Flow Job Shop	Job Shop			
	Disconnected Line Flow Batch		Batch		
	Connected Line Flow Assembly Line			Assembly Line	
	Continuous Flow Continuous				Continuous

Operations processes: Service systems

- Professional services
- Service shops
- Mass services

SERVICE PROCESS MATRIX

Enter your sub headline here



Facility layout

- Layout planning
 - Departments on site
 - Within departments
 - Workplaces
- Best configuration of resources::
 - Equipment
 - Infrastructure
 - People



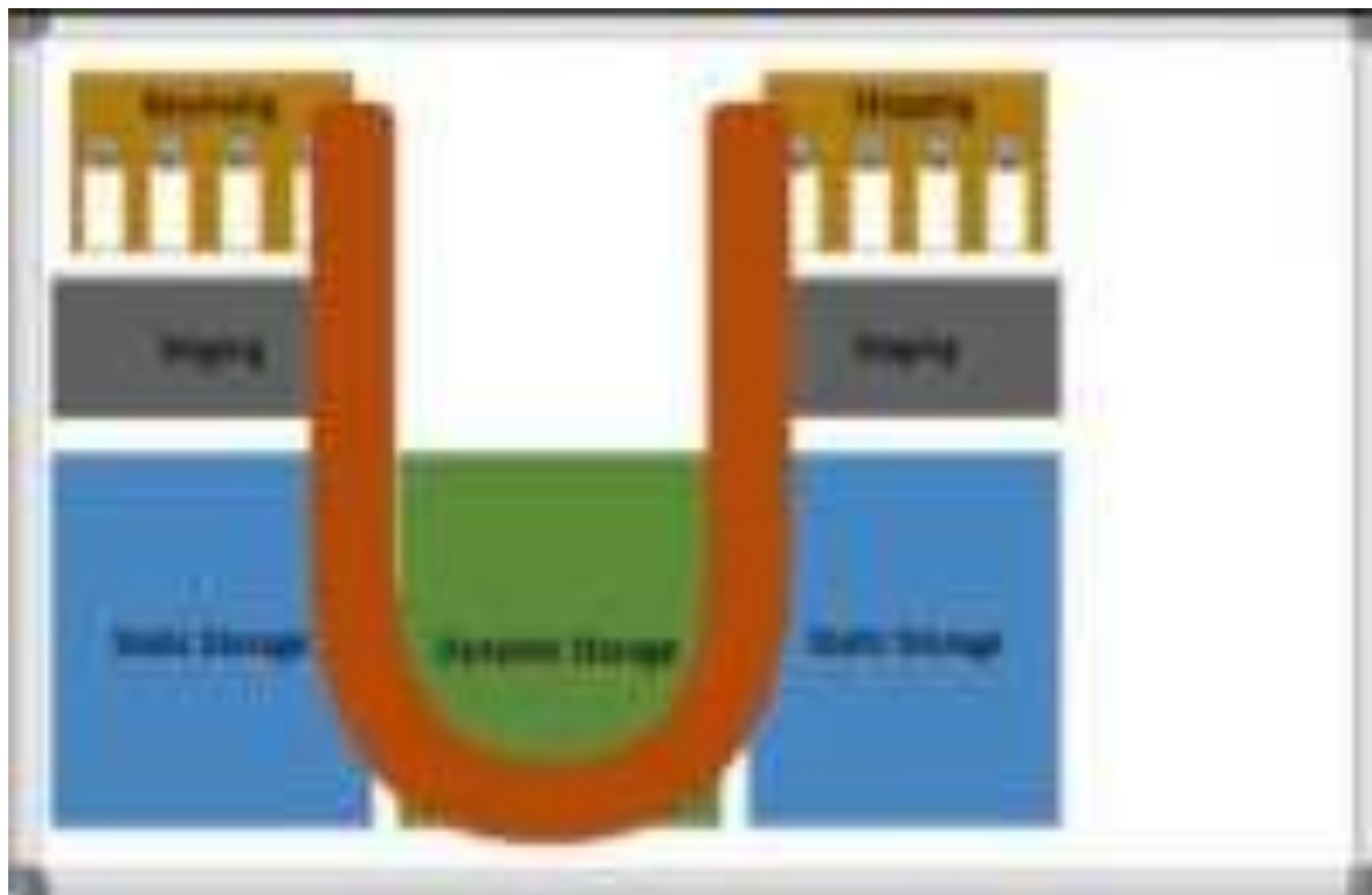
- Efficiency

Considerations

- Amount of movement of materials
- Health and Safety of staff and customers
- Levels of congestion
- Bottlenecks
- Utilisation of space, labour and equipment
- Level of inventory

Service

- Maximum product exposure
- Ambience of physical surrounding
- Customer perceptions of waiting times



Task | 10 Minutes

Design the most efficient festival layout



Timers And More

20:00

Quality

- **Define and understand precisely what the customer is expecting** and set up operations to deliver that
- 6 features to help to define quality in terms of what customers want:
 1. Functionality
 2. Performance
 3. Reliability
 4. Durability
 5. Customisation
 6. Appearance



Additional dimensions of service quality

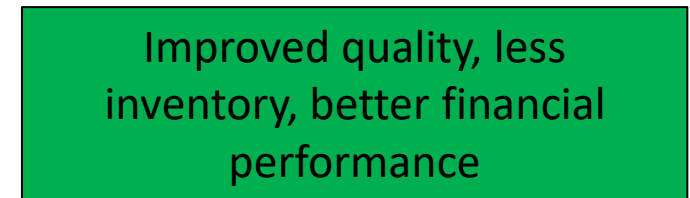
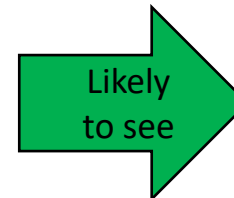
- Responsiveness
- Assurance (inspire confidence)
- Empathy (understanding and attentiveness)



Quality systems and procedures

- “Four sets of management practices shown to have enhance manufacturing and service quality” (Kaynak 2003)

1. Quality data and reporting
2. Supplier quality management
3. Product/service design
4. Process management



The evolution of quality management

Modern ops management evolved from 1920s

- Quality inspection Quality control



- Quality assurance



- Total quality management (TQM)



- Six Sigma

Table 13.1 Main tasks of quality management stages

Level of quality management	Characteristic actions of this level
Inspection – retrospective checking for quality after tasks have been completed. This is mainly done just by quality inspectors	• Salvage • Sorting, grading, re-blending • Corrective actions • Identify sources of non-conformance • Limited to a small number of police-like inspectors
Quality control (QC) – a system to detect quality problems as they occur, normally limited to operations	• Develop quality manual • Process performance data • Self-inspection • Product testing • Basic quality planning • Use of basic statistics • Paperwork controls
Quality assurance (QA) – a systems that is proactive with some emphasis on improvement, normally limited to operations and directly related areas	• Quality systems development • Advanced quality planning • Use of quality costs • Failure mode and effect analysis (FMEA) • Statistical process control (SPC) • Use of quality as a unique selling point for marketing
Total quality management (TQM) – a proactive approach involving everyone in an organisation.	• Policy deployment • Involve suppliers and customers • Involve all operations • Process management • Performance measurement • Teamwork • Employee involvement throughout the organization
Six Sigma – a proactive approach involving everyone in an organization, which has low failure rates	• Linkage to personal performance reviews and career progression • Emphasis on the reduction of errors • Application of Six Sigma tools to reduce process variation

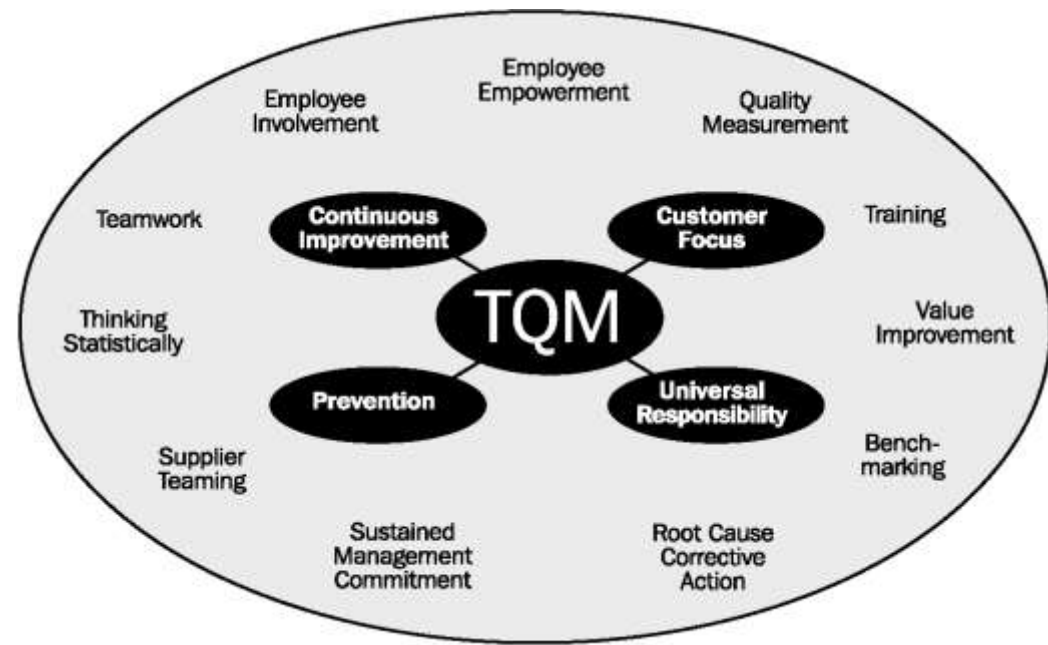
Source: adapted from Dale *et al.* (2007, p. 24).

Quality Management Stages

Quality management

- Total Quality Management:
 - Customer focus (meet their needs)
 - Comprehensive (all parts of the organisation)
 - Inclusive (includes everyone in organisation and supply chain)
 - Measurement (Tracking failures and getting things right first time)
 - Methods (clear systems and procedures)
 - Continuous (continuous improvement)
- Lean manufacturing

<https://www.youtube.com/watch?v=F5vtCRFRAK0>



Source: Berk and Berk (1993) p. 8

TQM systems

Improvements

- Fewer defects
- Lower inventory levels
- Reduced lead times
- Higher flexibility
- Increased employee satisfaction

Challenges

- A lack of company wide definition of quality
- A lack of formalised strategic plan for change
- A lack of customer focus, poor inter-organisational communication
- A lack of real employee empowerment
- Lack of trust in senior management
- View of quality as a quick fix
- Drive for short term financial results
- A lack of strong motivation
- Lack of leadership

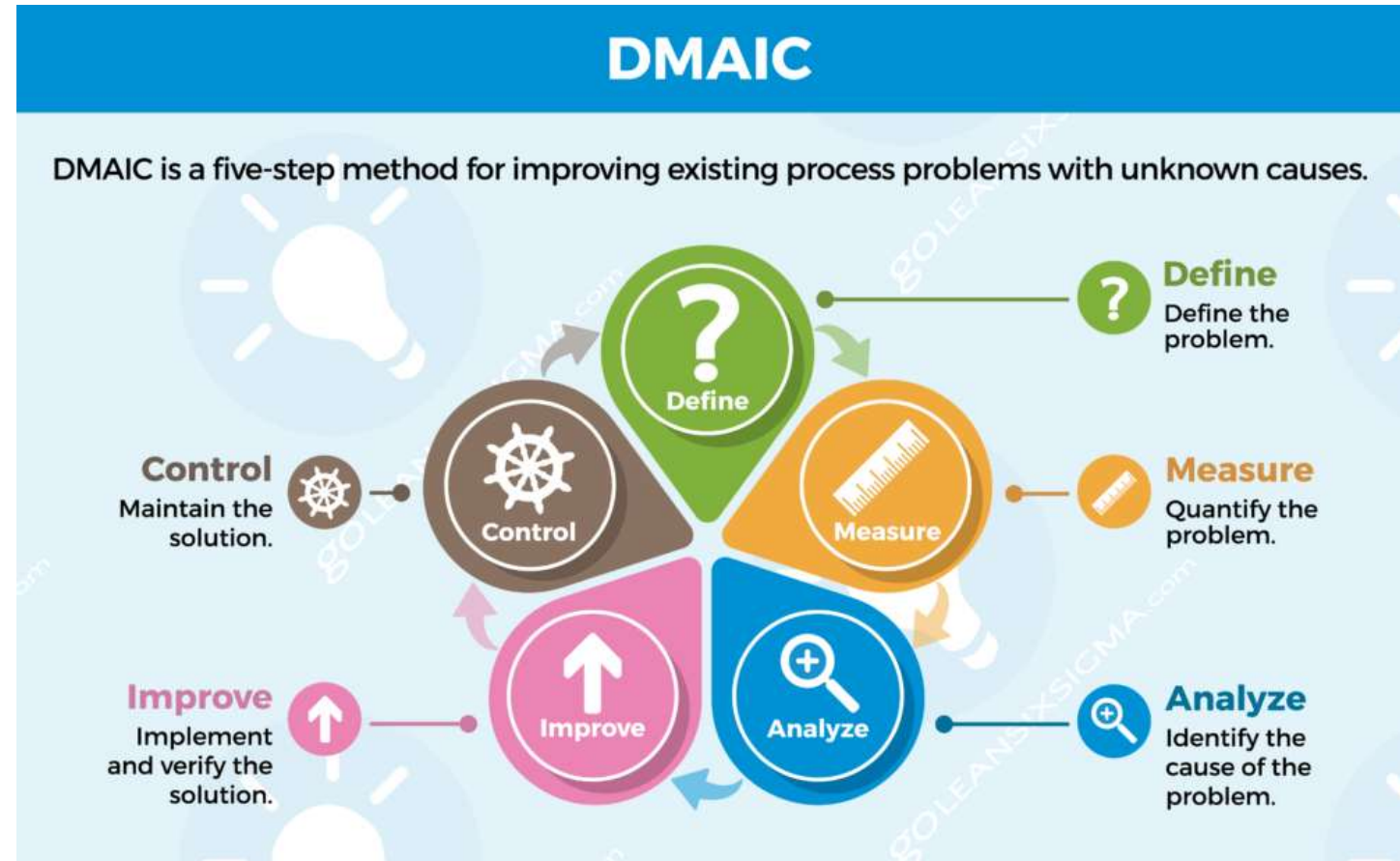
Six Sigma

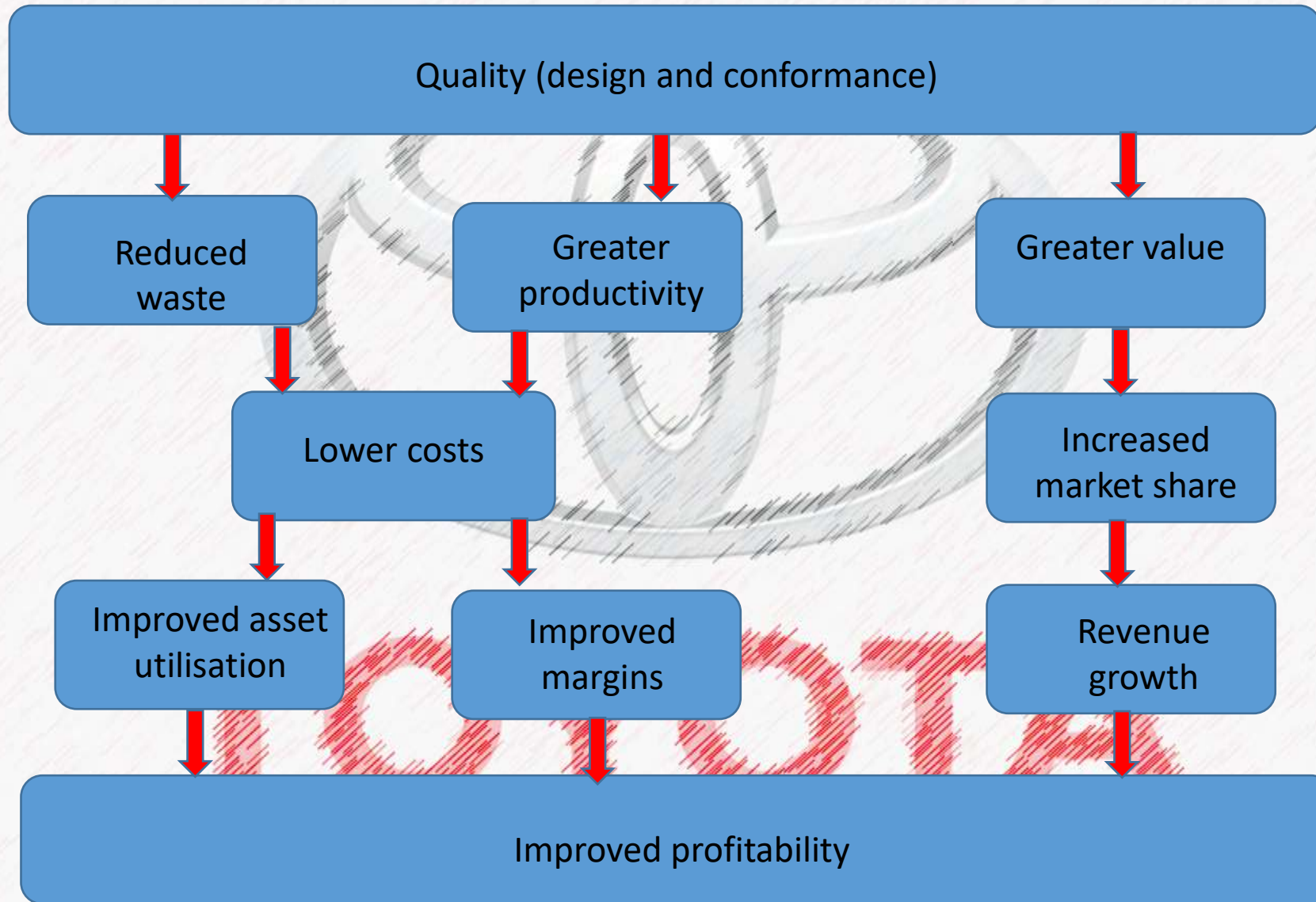
Quality concept with the goal of value creation through quality improvement

- Many objectives are similar to TQM (customer satisfaction, continuous improvement, team based activities and problem solving methodologies)
- It applies statistical methods to achieve no more than 3.4 defects per million opportunities in any process ,product or service
- Lean Six Sigma is an integrated approach of combining six sigma with lean production methodologies to improve process capability and eliminate waste

Six Sigma

- Continuous improvement or Kaizen methodology based on the DMAIC cycle
- Needs support from all levels of the organisation
- Needs strong leadership
- Employees are known as:
 - Green belts (3 years experience)
 - Black belt (experts)
 - Master black belts (ultimate experts who can train green and black belts)
- Training and certification





(Paton, Clegg et al 2011)

Case study – How Toyota lost its way

Management accountants are well placed to advise on the benefits of pursuing 'lean' manufacturing techniques, but some might argue that the credibility of those techniques has been undermined by Toyota's recent quality problems. Rather, the conclusion seems to be that commercial pressures diverted Toyota from the lean principles they formerly valued.

The lean manufacturing approach pioneered by Toyota concentrated on the identification and minimisation of waste in production, supply chains and management processes. Seven types of waste were targeted, including over processing, waiting times, and unnecessary transportation; so the approach was credited with improving quality, speeding up production and reducing costs. Strategies included getting suppliers to design cheaper and lighter components, and making productivity gains.

Toyota's recent growth has been significant. It has added 17 new manufacturing sites to the 58 it had in 2000, and expanded the product range significantly in a bid to rapidly increase market share. In 2000, Toyota produced 5.2m cars; by 2009 its capacity was 10m. Yet Toyota was able to cut more than \$10bn from global operating costs in the period 2000-2006.

The relentless pursuit of growth, the rapid expansion of supply chains, the pressure for ever-more productivity gains all combined to apparently undercut the commitment to quality which used to be part of the Toyota tradition. *'The company was hijacked by financially oriented pirates'* commented former top US executive Jim Press who left Toyota in 2007.

In 2009/2010 the quality problems came to a head when Toyota had to recall 5.3 million vehicles in the US, in relation to five separate issues affecting various models: accelerator entrapment by floor mats; unintended acceleration caused by 'sticky' pedals; issues with software relating to braking systems, power steering, and front drive shafts. US regulators linked the problems to 51 deaths, and had reports of unintended acceleration dating from 2003.

How could this happen in an organisation with a supposed quality oriented philosophy and an investigative approach which advocated going to the source of any issue? How did defective parts enter the supply chain? Why were quality problems not identified or actioned more promptly?

The consensus among commentators appears to be not that Toyota abandoned its lean approach; but that its rapid growth encouraged the company to follow the practices mechanically, rather than in the spirit of the principles. For example, the problem with the floor mats was not due to defective components from suppliers (as those met specifications); therefore it was not an engineering issue, but a design issue. This suggests that the suppliers were not working in the collaborative manner with Toyota assumed by a lean manufacturing approach, probably because the supplier network expanded so quickly. In a similar fashion, it seems likely that colleagues from the new plants did not benefit from Toyota's previous resource intensive policy of mentoring to share good practice and disseminate values.

Toyota's own assessment of where it went wrong was *'the pace at which we have grown may have been too quick... Toyota's priority has traditionally been first: safety; second: quality and third: volume. These priorities became confused... We pursued growth over the speed at which we were able to develop our products and our organisation'*. The company also felt it could have handled its response to the safety issues more effectively. *'We must think more from a customer first perspective rather than a technical perspective when investigating complaints..we must communicate faster, better and more effectively with customers and regulators'*.

Toyota's solutions include new regional quality officers to give regions more autonomy and decision making with regard to recalls and other safety issues. It also established a new special committee for global quality, and teams to investigate quickly reports of unintended acceleration. It reinforced its design capability by transferring one thousand engineers to focus on design of components and other quality issues. It lengthened product lifecycles by four weeks to give more time to address safety/quality issues and improved monitoring systems to better capture intelligence on quality/safety concerns from various sources (including web mentions, customer calls and government databases). It intends to equip more vehicles with the technology to diagnose problems and report faults ('black boxes') and has allowed external parties to review its technology.

Case Studies

Let's look at a few case studies of successful quality management methodology in practice....

<https://asq.org/quality-resources/articles/case-studies>

The Cost of Quality

- Not cheap –2001 General Electric spent \$11-16 billion to implement quality management strategy (www.isixsigma.com)
1. Prevention costs - quality planning ,training
 2. Appraisal (degree of conformance) costs - inspection, audits
 3. Failure costs
 - External – warranty, returns , penalties
 - Internal - scrap, rework, re-inspection

Quality management - considerations

- Defining quality
- Importance of quality management for organisations
- Evolution of quality management
- Quality journey – to strive to practice better quality management and raise standards
- Operational tools to assess and improve quality –TQM, Lean, Six Sigma
- Contemporary view is that it is always good to invest in quality
- Emerging issue of corporate social responsibility in assessing quality

The 8-Step Process for Leading Change - Kotter



<https://www.kotterinc.com/8-steps-process-for-leading-change/>

Food for Thought????



References

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- Sprague, I. (2007) *Evolution of the field of operations management*, Journal of operations management, vol 25, no2, pp 219-38